PIC OF X IS THE GALOIS-INVARIANT PIC OF X_{K'}

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1558-1568 of sources/stacks-pic/pinned-original-source.tex; proof span is lines 1570-1623, the complete author proof (52 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 5 of queue_pic.json (pb=2833, stmt 1558-1568, proof 1570-1623).

Editorial note. stacks-pic wave-1 (pi01): fresh adjudication accept (2026-10-14); pool per standing queue hypercovering -> pic (SMC reassigned to Zhen per Chief 08:5x); pic.tex 27 labels/23 with proof judged 21 accept / 2 reject / 4 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0CDR: sources/stacks-pic/pinned-original-source.tex, source lines 1623–1623. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BU2: sources/stacks-pic/pinned-original-source.tex, source lines 1623–1623. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B8W: sources/stacks-pic/pinned-original-source.tex, source lines 1623–1623. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01YV: sources/stacks-pic/pinned-original-source.tex, source lines 1623–1623. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CC5: sources/stacks-pic/pinned-original-source.tex, source lines 1623–1623. Cited by the retained statement or proof but not part of

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the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let k be a field. Let X be a quasi-compact and quasi-separated scheme over k with $H^0(X, \mathcal{O}_X) = k$. If X has a k-rational point, then for any Galois extension k'/k we have*

$$\mathrm{Pic}(X) = \mathrm{Pic}(X_{k'})^{Gal(k'/k)}$$

Moreover the action of $Gal(k'/k)$ on $\mathrm{Pic}(X_{k'})$ is continuous.

Proof. Since $\mathrm{Gal}(k'/k) = \mathrm{Aut}(k'/k)$ it acts (from the right) on $\mathrm{Spec}(k')$, hence it acts (from the right) on $X_{k'} = X \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k')$, and since $\mathrm{Pic}(-)$ is a contravariant functor, it acts (from the left) on $\mathrm{Pic}(X_{k'})$. If k'/k is an infinite Galois extension, then we write $k' = \mathrm{colim}\, k'_\lambda$ as a filtered colimit of finite Galois extensions, see Fields, Lemma 0BU2. Then $X_{k'} = \lim X_{k_\lambda}$ (as in Limits, Section 01YV) and we obtain

$$\mathrm{Pic}(X_{k'}) = \mathrm{colim}\, \mathrm{Pic}(X_{k_\lambda})$$

by Limits, Lemma 0B8W. Moreover, the transition maps in this system of abelian groups are injective by Varieties, Lemma 0CC5. It follows that every element of $\mathrm{Pic}(X_{k'})$ is fixed by one of the open subgroups $\mathrm{Gal}(k'/k_\lambda)$, which exactly means that the action is continuous. Injectivity of the transition maps implies that it suffices to prove the statement on fixed points in the case that k'/k is finite Galois.

Assume k'/k is finite Galois with Galois group $G = \mathrm{Gal}(k'/k)$. Let $\mathcal{L}$ be an element of $\mathrm{Pic}(X_{k'})$ fixed by G. We will use Galois descent (Descent, Lemma 0CDR) to prove that $\mathcal{L}$ is the pullback of an invertible sheaf on X. Recall that $f_\sigma = \mathrm{id}_X \times \mathrm{Spec}(\sigma) : X_{k'} \to X_{k'}$ and that σ acts on $\mathrm{Pic}(X_{k'})$ by pulling back by f_σ. Hence for each $\sigma \in G$ we can choose an isomorphism $\varphi_\sigma : \mathcal{L} \to f_\sigma^*\mathcal{L}$ because $\mathcal{L}$ is a fixed by the G-action. The trouble is that we don't know if we can choose φ_σ such that the cocycle condition $\varphi_{\sigma\tau} = f_\sigma^*\varphi_\tau \circ \varphi_\sigma$ holds. To see that this is possible we use that X has a k-rational point $x \in X(k)$. Of course, x similarly determines a k'-rational point $x' \in X_{k'}$ which is fixed by f_σ for all σ. Pick a nonzero element s in the fibre of $\mathcal{L}$ at x'; the fibre is the 1-dimensional $k' = \kappa(x')$-vector space

$$\mathcal{L}_{x'} \otimes_{\mathcal{O}_{X_{k'}, x'}} \kappa(x').$$

Then $f_\sigma^* s$ is a nonzero element of the fibre of $f_\sigma^*\mathcal{L}$ at x'. Since we can multiply φ_σ by an element of $(k')^*$ we may assume that φ_σ sends s to $f_\sigma^* s$. Then we see that both $\varphi_{\sigma\tau}$ and $f_\sigma^*\varphi_\tau \circ \varphi_\sigma$ send s to $f_{\sigma\tau}^* s = f_\tau^* f_\sigma^* s$. Since $H^0(X_{k'}, \mathcal{O}_{X_{k'}}) = k'$ these two isomorphisms have to be the same (as one is a global unit times the other and they agree in x') and the proof is complete. □